

Deep Punj, PhD

Physics-Informed Data Scientist & AI Specialist

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[PLACEHOLDER résumé — replace with the real one before sharing]

SUMMARY

Physics-trained data scientist turning conservation laws and governing equations into differentiable, deployable ML systems. Expert in physics-informed neural networks (PINNs), surrogate modeling, and production forecasting; ships models end-to-end with rigorous MLOps.

SKILLS

Languages & Data: Python (expert), SQL, NumPy/Pandas, Polars

Deep Learning: PyTorch, JAX, scikit-learn

Interfaces & Apps: Streamlit, FastAPI, Plotly

MLOps & Infra: Docker, MLflow, GitHub Actions

EXPERIENCE

Senior Data Scientist — [Company placeholder]

2021 – present · Placeholder location

- Built physics-informed surrogate models that replaced expensive CFD runs in design loops, cutting evaluation time by orders of magnitude. [placeholder metric]
- Deployed forecasting models to production with automated retraining, monitoring, and drift checks. [placeholder]
- Led cross-functional rollout of model governance and reproducibility standards. [placeholder]

Research Scientist / PhD Researcher — [Institution placeholder]

2018 – 2021 · Placeholder location

- Developed differentiable PDE solvers and applied PINNs to flow and heat-transfer problems. [placeholder]
- Authored 12+ peer-reviewed publications. [placeholder count]

SELECTED PROJECTS

- **PINN Flow Solver** — Deep network solving Navier–Stokes with embedded conservation laws. [placeholder]
- **Heat-Exchanger Surrogate** — Fast thermal surrogate for real-time design optimization. [placeholder]
- **Grid Load Forecasting** — Production forecasting service with physics-informed constraints. [placeholder]

EDUCATION

PhD, Physics — [University placeholder]

Thesis: Physics-informed machine learning for physical systems. [placeholder]

MSc, Applied Physics — [University placeholder]